

Business Performance Analysis and Service Optimization: A Primary Data Study of a Professional Laundry Business.

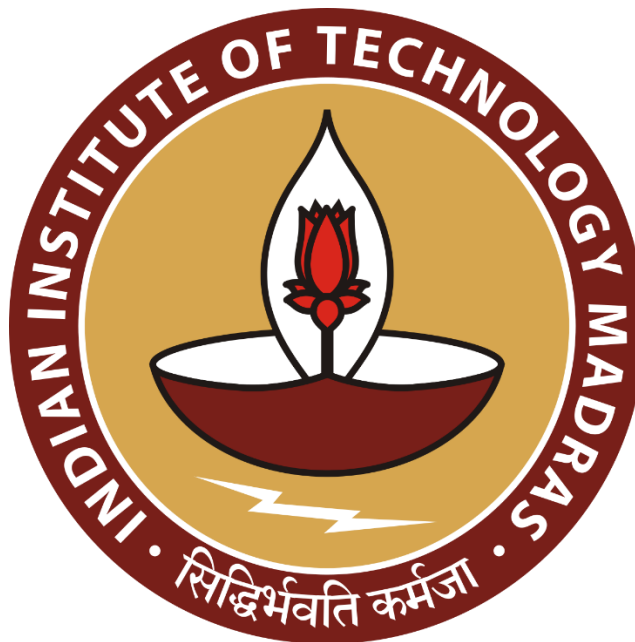
A MID-TERM REPORT FOR THE BDM CAPSTONE PROJECT

Submitted by:

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(BS) DEGREE IN DATA SCIENCE AND APPLICATIONS

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1. Executive Summary

RDS Associates operating under the trade name ‘Rinz n Dry’ is a privately owned dry-cleaning and laundry service business. It is an independent B2C business organization operating since 2022, located at MM Towers, Fatima Nagar, East Fort, Thrissur, Kerala. This Business provides multiple garment services like wash and iron, wash and fold, dry cleaning for various types of garments, shoe cleaning, bag cleaning, curtain cleaning, and blanket washing with pickup and delivery.

As a part of the project, a primary data set of 2,607 customer order records was collected directly from the business, covering the period from 10 March 2026 to 10 July 2026 – over a period of four months across 106 working days. The dataset was organized and cleaned using Excel and Python (Pandas), and statistical analysis was performed using Stata and Excel.

The analysis addresses three problem statements stated in the project proposal: (i) revenue and service demand imbalance, (ii) operational efficiency and turnaround time (TAT), and (iii) customer retention and behavioural patterns. Descriptive statistics, correlation analysis with significance testing, hypothesis tests (t-test, chi-square, ANOVA), and a preliminary weekly time-series trend were used to investigate these problems.

The analysis shows that Wash & Iron is the dominant service, which makes up 47.8% of all orders (1246 of 2607) and 62.9% of total revenue (Rs. 8,24,685 of Rs. 13,11,105). This shows that Wash & Iron is an important factor for capacity and resource planning. Turnaround time is primarily determined by the service policy, not order size. Every Normal-service is completed in exactly 4 days, including Sundays (7 days for Blanket Washing and Shoe Cleaning) and every Express order is completed in exactly 1 day, with zero variance within each group; confirming that TAT is a scheduling/policy variable, not something that drifts with garment count.

Regular customers are the core of the business. 46.4% of customers (487 of 1,050) have placed 2 or more orders, and this group contributes 78.4% of all orders and 79% of revenue (approx. Rs. 10,36,195 of Rs. 13,11,105). A two-sample t-test shows no statistically significant difference in average spending per order between repeat and single-order customers (Rs. 507 vs Rs. 488, $p = 0.166$) - the higher revenue contribution is by purchase frequency, not by a larger individual transaction.

Customer complaints were recorded for 14.4% of orders (376 of 2,607), of these, 90.7% were resolved through rectification, compensation or refund, while 9.3% remained pending or escalated, giving a clear benchmark for service performance. Weekly revenue is almost stable over all weeks with an average of Rs. 74,672/week, a coefficient of variation of 7.6%, a near-flat linear trend of about Rs. 92 per week. Monthly revenue per working day is also stable on March to July (Rs. 12,311 to Rs. 12,721) indicating every month has roughly same amount of business.

The final report will extend these findings and provide a complete report with actionable recommendations for RDS Associates to improve the business.

2. Proof of Originality of the Data

Letter from the Organization

Date: 18/06/2026

To Whom It May Concern,

This is to certify that **RDS Associates**, operating under the trade name **Rinz n Dry** has granted permission to **Mr. Arun Sankar E K** (Roll No: **21F3002256**), a student of the **BS Degree in Data Science and Applications**, **IIT Madras**, to collect and use data from our organization for his academic project titled:

"Business Performance Analysis and Service Optimization: A Primary Data Study of a Professional Laundry and Dry-Cleaning Business."

We understand that the data collected from our organization will be used **only for academic and research purposes** as part of the above-mentioned project. We also understand that the data will be analysed solely for educational purposes and will not be disclosed, shared, or distributed to any third party without our permission.

The student has assured us that the confidentiality of our business information will be maintained and that any findings presented in the project will be used only to fulfil the academic requirements of the IIT Madras BS Degree Programme.

We have no objection to the student using the required business data for this project.

Sincerely,

Shanitha
RDS Associates



Note: Rinz n Dry is the trade name under which the business operates, while RDS Associates is its registered legal/GST entity. Both names refer to the same organization.

As part of the proof of originality for the collected data, I am providing the following supporting documents and materials: Letter from the Organization, Images related to the organization, interaction video with the business owner, working dataset are attached the below drive link.

https://drive.google.com/drive/folders/1YtPg76c2nBLU457QleDCDBHk0rA_0rs1?usp=sharing

3. Metadata and Descriptive Statistics

3.1 Data Source and Structure

The primary data set "rds_df", contains 2,607 customer order records collected directly from the organisation. The dataset covers the period from 10 March 2026 to 10 July 2026, (106 working days, excluding Sunday). This Dataset contains 16 columns and includes information on order timing, customer type, service characteristics, billing, delivery, TAT and all (SI no used as identifier) the variables are used for analysis.

Column	Unit / Type	Description	Linked PS
Sl No	Integer	Serial number of the order record	-
Order Date	Date	Date on which order was placed	PS-2
Order Time	24-hr time	Time at which order was placed	PS-2
Customer ID	Integer	Unique identifier of customer	PS-3
Customer Type	Categorical	‘New’ (first visit) or ‘Regular’ (2nd visit onwards)	PS-3
Type of Service	Categorical	Service selected by customer	PS-1
Service Level	Categorical	Normal or Express processing	PS-2
No. of Garments	Integer	Number of garments in order	PS-1, PS-2
Bill Amount	Integer	Amount charged for the order	PS-1, PS-3
Payment Mode	Categorical	UPI / Cash / Card	PS-3
Pickup/Delivery Type	Categorical	Walk-in / Free / Chargeable delivery	PS-1, PS-3
Extra Distance	km	Distance beyond the free 3.5 km radius	PS-2
Extra Distance Rate	Rs. per km	Rate charged for additional distance	PS-2
Delivery Date	Date	Date on which order was delivered	PS-2
Turnaround Time	Integer	Time between order and delivery	PS-2
Complaint Status	Categorical	Blank / Pending / Escalated / 3 Resolved types	PS-2

Table 1: Dataset metadata, units, and linkage to the three project problem statements

No missing values were identified in the core operational variables such as Customer ID, Type of Service, Bill Amount, and date columns. The Extra Distance and Extra Distance Rate fields are applicable only to Chargeable Pickup/Delivery orders and filled with 274 observations; so, blank values in these fields indicate that the variable not applicable not a missing data. Similarly, blank values in Complaint Status indicate that no complaint was recorded. Complaints were recorded for 376 of 2,607 orders (14.4%). The dataset contains records only for Monday–Saturday, consistent with the business operating schedule.

Also, the variables are linked directly to the 3 problems statements, builds all subsequent analysis for the study.

PS1 - Revenue and Service Demand Imbalance; examines which services are generating most revenue and how it’s related to the order volume? **PS2 - Operational Efficiency and Turnaround Time;** examines TAT, peak ordering time and how it can be managed? **PS 3 - Customer Retention and Behavioural Patterns;** examines customer loyalty, repeat purchasing behaviour and does that reproduce disproportionate profit?

3.2 Descriptive Statistics

3.2.1 Numerical Variables

Variable	Count	Mean	Std. Dev.	Min	25%	50%	75%	Max	CV
No. of Garments	2606	3.64	2.16	1	2	3	5	8	59.20
Bill Amount (Rs.)	2606	502.92	289.83	90	280	440	680	1650	57.63
Extra Distance (km)	274	3.25	1.53	0.5	2.1	3.3	4.5	6	47.09
Extra Distance Rate (Rs/km)	274	7.47	0.50	7	7	7	8	8	6.69
Turnaround Time (Days)	2606	3.85	1.26	1	4	4	4	7	32.67

Table 2: Descriptive statistics of key numerical variables

Both no of Garments (CV ~ 59%) and Bill Amount (CV ~ 58%), shows relatively high variation, reflecting differences in order size, service type and garment composition. This variation is further examined in §3.2.2 and differ from prior assumption. TAT has a CV of 32.67%, indicating considerable relative variation. §3.2.2 and §4.1 shows, it is completely described by service scale and, within normal, by the difference in the service type, not by the changes in the order volume.

3.2.2 Service-wise Order and Revenue Distribution

Services	Orders	Order Share	Revenue (Rs.)	Rev. Share	Avg. Bill (Rs.)	Avg. TAT (days)
Wash & Iron	1,246	47.8%	8,24,685	62.9%	662	3.63
Wash & Fold	334	12.8%	99,365	7.6%	298	3.72
Blanket Washing	97	3.7%	68,825	5.3%	710	6.32
DC - Shirt/Fork/T-shirt	206	7.9%	61,565	4.7%	299	3.62
DC - Pant/Kurtha	202	7.8%	59,280	4.5%	293	3.66
Saree/Mundu - Dry Clean	120	4.6%	52,665	4.0%	439	3.75
Shoe Cleaning	89	3.4%	42,765	3.3%	481	6.39
Curtain Cleaning	101	3.9%	29,115	2.2%	288	3.76
Jacket Dry Clean	61	2.3%	27,720	2.1%	454	3.51
Bag Cleaning	74	2.8%	27,565	2.1%	373	3.68
Saree/Mundu - Normal Wash	77	3.0%	17,555	1.3%	228	3.69

Table 3: Service-wise order volume, revenue, and turnaround time (10 Mar – 10 Jul 2026)

Revenue share of Wash & Iron is 62.9%. This is exceeding its volume share which is only 47.85% of the business. At the same time this service is also carries greater average bill amount Rs 662 than other higher order size services. All these making this service dominant in both cases of order volume and revenue contributor. The blanket washing service has differently pre-dominated. This service is dominated by highest average bill of Rs 710 and in addition with Shoe cleaning, it is showing the highest Turnaround time of 6.3-6.4 days in average.

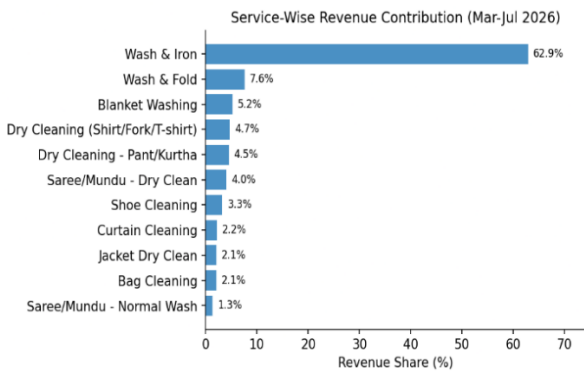


Fig 1: Service-wise revenue contribution, March–July 2026

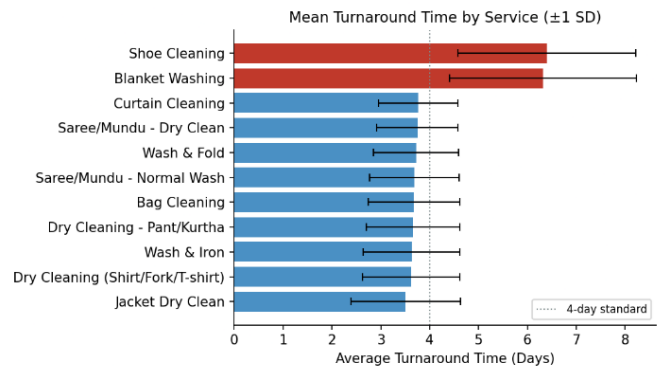


Fig 2: Mean TAT by service (± 1 SD), with 4-day standard marked

3.2.3 Categorical Variable and Monthly Revenue

The payment-mode analysis shows that UPI is the most commonly used payment method (52.9%), followed by Cash (37.4%) and Card (9.7%) - showing digitally dominated customer base. However, payment mode is not significantly associated with customer type ($\chi^2 = 0.19$, $p = 0.91$), indicating that payment preference does not distinguish Regular customers from New customers in this dataset.

Coming to the pickup and delivery type, 45.1% of orders are walk-in, 44.3% are Free Pickup and Delivery, and 10.5% use Chargeable Pickup/Delivery. A one-way ANOVA test shows that, there is no statistically significant difference in average Bill amount and Delivery Type ($F = 0.31$, $p = 0.73$; average bills range narrowly from Rs. 498 to Rs. 508 across the three types). Which shows that delivery type determines on the basis of distance and convenience, not on the order size and bill amount.

Month	Working Days	Orders	Revenue (Rs.)	Revenue / Working Day (Rs.)
March 2026 (from 10th)	19	459	2,33,950	12,313
April 2026	26	646	3,20,085	12,311
May 2026	26	634	3,21,885	12,380
June 2026	26	641	3,30,750	12,721
July 2026 (to 10th)	9	227	1,04,435	11,604

Table 4: Monthly orders and revenue, normalised by working days

When the revenue is considering per working day, Rs. 12,300 – Rs 12,720 is the range for every month from March to June with the spread of 3%. For the month of July, the data collected only for 9 working days hence not decisively analyse. There is no substantially busier month in the 4 month dataset. Hence within the dataset, it is only appropriate to state the month to month variation rather than substantiating seasonal pattern.

4. Detailed Explanation of Analysis Process and Methodology

4.1 Methodology and Analytical Approach

The study primarily focuses on the descriptive statistics, because no comparison across services, customer groups, or time periods is meaningful until the overall structure of the data is understood. Descriptive statistics and service level tabulation (mean, median, SD, range, CV) is used to create the structural baseline of order size, billing and TAT. This is required for any comparison within the different services to be meaningful.

Hypothesis tests (t-test, chi-square, ANOVA) were then used to validate whether apparent differences - e.g., ‘repeat customers spend more’, ‘delivery orders have higher bills’- were statistically significant or due to chance, before being reported as findings.

Correlation analysis tested whether operational variables (garment count, order hour) actually explained the observed variation in bill amount and TAT; results are reported as associations, not causal mechanisms. Weekly time-series analysis provided an exploratory view of revenue stability once four complete months of data were available.

Further the study utilized the Pearson Correlation method to identify the association between various operational variables to allocate the recourses more efficiently. Correlation values are treated as associational relation, bit not as a causal effects. §4.1, this is where the service lever segregation reveals the real mechanism behind the correlation. The exploratory view of revenue stability of the business is given by the weekly time series trend on 16 study. Because this contains only 16 points for the complete weeks which is essentially low points from the ARIMA test.

4.1.1 Correlation Analysis — Association, Not Causation

The correlation analysis covers four numerical variables, with statistical significance (p-value) reported for each pair so that only associations unlikely to be due to chance are discussed.

Variable Pair	r	p-value	Interpretation
No. of Garments, Bill Amount	0.785	< 0.001	Strong, significant — order size is the main driver of bill value
No. of Garments, TAT	-0.165	< 0.001	Weak, significant — larger orders do not take meaningfully longer
Bill Amount, TAT	-0.192	< 0.001	Weak, significant — higher-bill orders are, if anything, slightly faster
Order Hour, Bill Amount	-0.009	0.643	Not significant — no time-of-day effect on order value
Order Hour, TAT	0.015	0.444	Not significant — no time-of-day effect on turnaround

Table 5: Correlation coefficients with significance tests

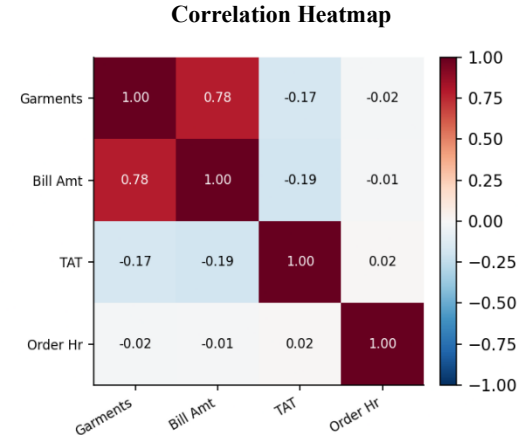


Fig 3: Correlation heatmap of key operational variables

Bill amount and garment count are strongly associated ($r = 0.785$), which is consistent with resources like chemicals, machine time, staff etc. should allocate primarily with the garment volume. There is only weak correlation between turnaround time and order size $r =$ under 1.19. Hence this is an evidence for saying the delay in Delivery is not depending on the order size. the cross tabulation of TAT and service type reveals, normal order is completed within in the standard TAT of 4 days while special services like Shoe cleaning, Blanket wash are taking up to 7 days. TAT is a deterministic policy outcome. Because the negative correlation with order size and bill amount is the after effect of the express delivery system not otherwise

4.1.2 Customer Segmentation – With statistical validation

In the dataset, the cells tagged with ‘New’ indicating customer’s first visit and ‘Regular’ is indicating customer is in his/her second visit onwards. The customers are grouped by a unique customer ID. Out of 1050 customers, 563 customers (53.6%) ordered only once in 4 months, while 487 customers (46.4%) are done their second visit between 2 and 14 orders each.

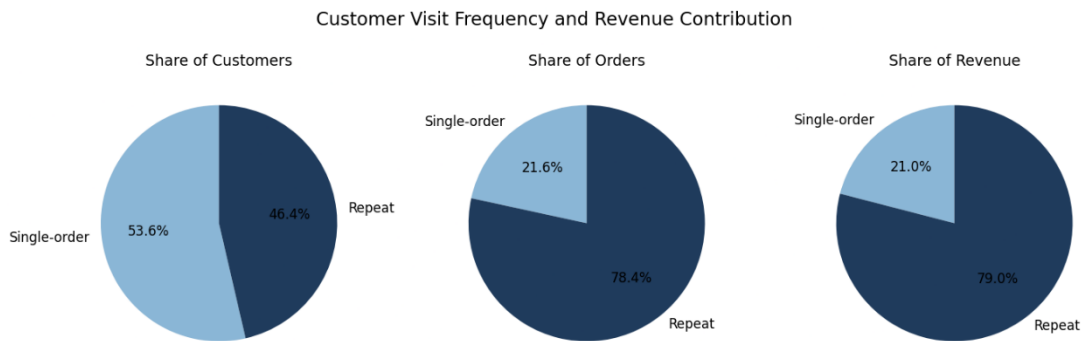


Figure 4: Customer visit frequency and revenue contribution, single-order vs repeat customers

The 78.4% of all orders and 79.05 of the revenue contribution are from these 487 repeat customers. A significant group that encourage to prioritizing retention of this repeat customer group over the one time visitors. There is no substantial difference between single and repeat order customers as per a two sample t test on bill amount. (Rs. 506 vs Rs. 488.3 per order, $t = 1.385$, $p = 0.166$): customer return is significant to show the revenue concentration, not by more spending from repeating customers per visit. Operationally, this distinction is very significant. The business should focus on converting the 53.6 % of customers to come again rather than promoting them to increase the order volume.

The basic retention metrics will be directly examined using Customer ID and Order Date. Across the 487 regular customers, 15 days is the median gap between customers first and second visit while mean gap is 21.0 days. And among all repeat customers, 11 days is the median gap between the two consecutive visits. Hence within two – three weeks, most of the regular customers visiting the shop again. The simple recency check in the 532 customers out of 1050 customers, give an order within in last 30 days of the 4 month window, but the regular customers made their last order before median of 22 days. This is not a full survival based model, but an activity measure only. For a survival model, this need a longer observation base to define trustable inactivity threshold.

4.1.3 Preliminary Time- Series Analysis of Weekly Revenue

For Preliminary Time- Series Analysis of Weekly Revenue, the study utilized 16 complete Monday to Saturday weeks.

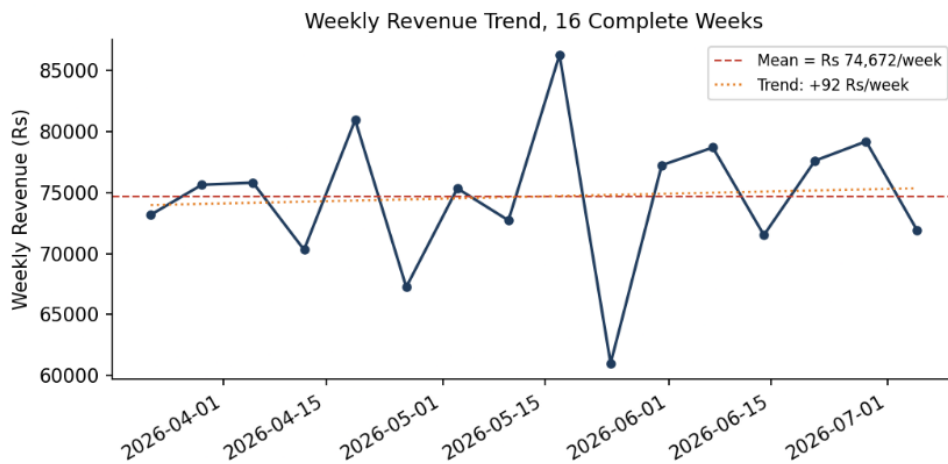


Figure 5: Weekly revenue trend, 10 March – 10 July 2026 (partial edge weeks excluded)

The study shows that, the revenue of 16 complete weeks are stable enough. The mean amount is Rs. 74,672, standard deviation CV 7.6% (Rs. 5,668), varying from Rs. 61,005 to 86.305 with no upward or downward for multiple weeks. The 16 weeks are showing a linear trend with slope of only +Rs. 92/week. This is signalling no substantial growth or decline between this weeks. With only 16 observational point a full ARIMA model will not perfectly fit here. Hence this weekly trend is an exploratory, descriptive model of stability only.

4.2 Detailed Exploratory Data Analysis

4.2.1 Demand Timing – Hour and Day of Week

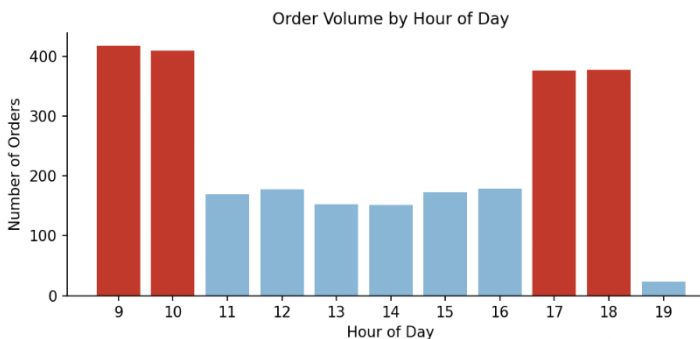


Figure 6: Order volume by hour of day

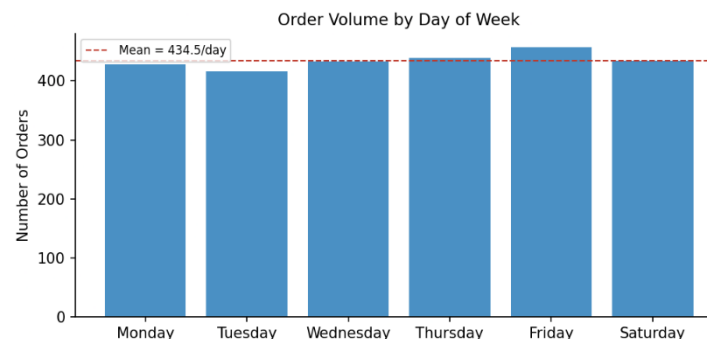


Figure 7: Order volume by day of week

The 60.6% of orders are placing between 9-10 AM and 5-6 PM together, around 1,580 of 2,607 orders. This is showing the most of the customers are placing their order on their way to job and collecting them while returning. although the dataset does not record customer occupation or commute behaviour, therefore, it is a valid inference of the timing pattern. The time between 11 AM and 5.00 PM are relatively low order placing time, only 150-180 per hour. Coming to the per day orders, a week showing stability in all days. Almost every day is showing 434.5 orders per day and comparatively Tuesday is the lowest day. Hence this is showing that the staffs should be concentrated to the peak hours rather than on specific peak days.

4.2.2 Service Quality – Complaint Rate and Resolution

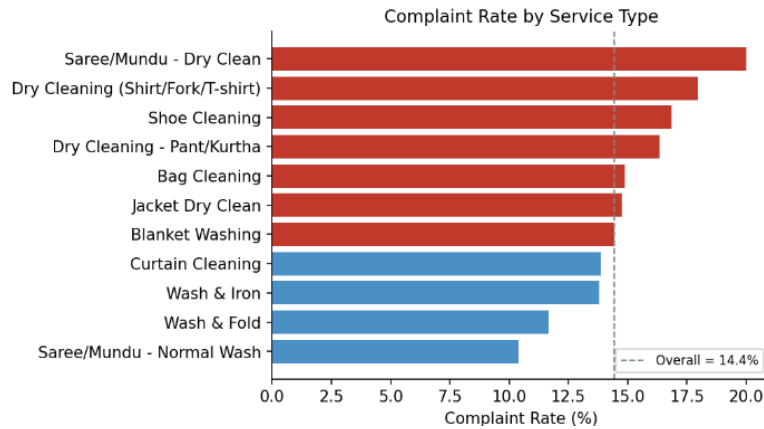


Figure 8: Complaint rate by service type

In the span of 4 months data, the business got only 376 complaints from 2,607 orders. Hence the complaint rate is 14.4%. As per the dataset, the shop has rectified 182 complaints over orders, 114 compensated, 45 refunded, 20 is still remaining and 15 are escalated. Most of the complaints are placing over Saree/Mundu Dry cleaning (20%) and Dry cleaning of shirt, fork, T –shirt (18%). Saree/Mundu normal Wash has 10% complaint rate and Wash and Fold has 11.7% complaint rate. Wash and Iron holds highest order volume and carries complaint rate of 13.8 %. This field generate largest actual number of complaints despite without being problematic. The most required attention should be go for the two dry cleaning services and there is a need for identifying the root cause of them which the current dataset is not observing.

6. Results and Findings

6.1 Problem Statement 1 — Revenue and Service Demand Imbalance

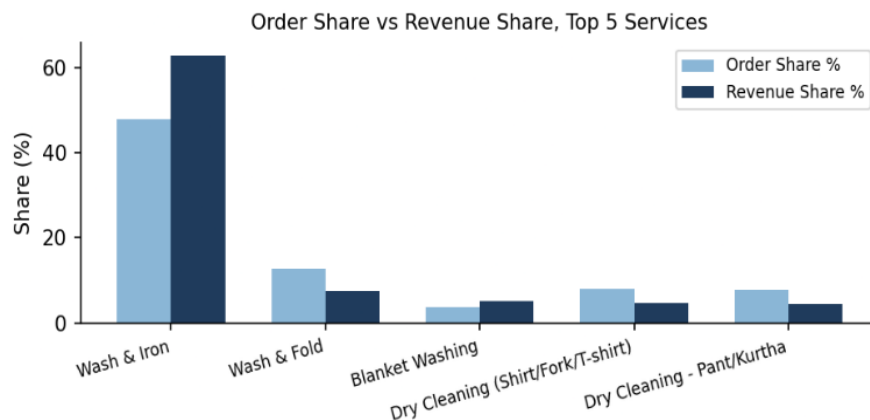


Figure 9: Order share vs revenue share, top 5 services by revenue

Wash & Iron is dominated both in volume and revenue. It contains 47.85% of orders and 62.9% of the revenue contributor. The revenue share is higher than volume share because the average bill share is also higher than dataset average for Wash and Iron service. In the case of premium-service promotion is only a partial solution. Blanket wash has the highest average bill (Rs.710) but also the longest TAT (6.3 days), so promoting it without first addressing turnaround time would likely convert demand into complaints rather than revenue.

Business Implication: The business should build the staff & capacity mainly around the wash and iron services due to higher volume and revenue contribution. At the same time promoting premium services to be coupled with capacity building, not addressing as a demand generation gap.

6.2 Problem Statement 2 — Operational Efficiency and Turnaround Time

The 60.6% of orders are placing between 9-10 AM and 5-6 PM together, around 1,580 of 2,607 orders. Coming to the per day orders, a week showing stability in all days. The TAT of Blanket Wash and Shoe cleaning is between 6.3 – 7 days. It is much higher than the standard TAT. Coming to the complaint rate, the overall complaint rate is 14.4%. Dry Clean (20.0%) and Dry Cleaning of Shirt/Fork/T-shirt (18.0%) are highest. There is only weak connection between the TAT and order size as per the correlation analysis ($r = \text{under } 0.19$).

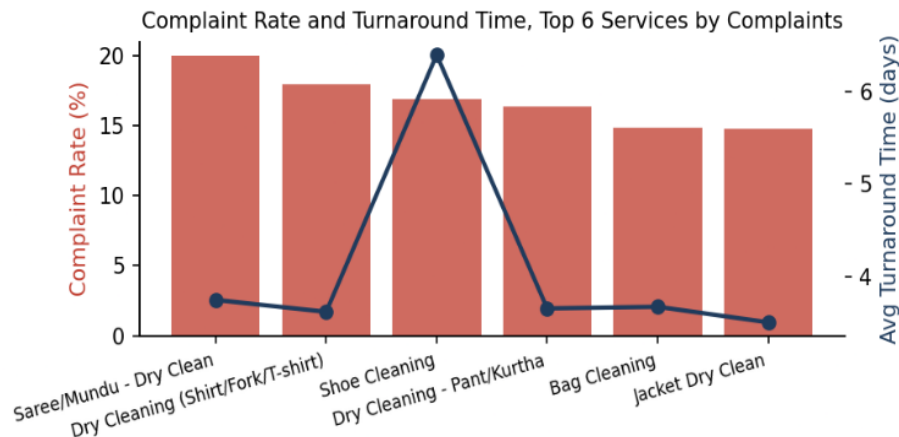


Figure 10: Complaint rate and average turnaround time, top 6 services by complaint rate

Business Implication: This is showing that the staffs should be concentrated to the peak hours rather than on specific peak days. Coming to TAT, this is a difference in the policy design, not the efficiency failure. The proper communication towards the customer about the TAT of specific services will drive more accuracy. The most required attention should be go for the two dry cleaning services and there is a need for identifying the root cause of them which the current dataset is not observing.

6.3 Problem Statement 3 — Customer Retention and Behavioural Patterns

487 of 1,050 (46.4%) customers are regular or repeated customers, however they are contributing 78.4% of orders and 79% of the revenue. Some results are showing average bill for the regular customer is Rs. 506.9 and Rs. 488.3 for the single order. Here the difference is not very significant ($t = 1.385$, $p = 0.166$).

Business Implication: Hence the business should focus on active repeat base customers rather than focusing on the small elite group. The business retention strategy should move from protecting few VIPs to translate the 53.6% one-time visitors to regular customers. And coming to the spending pattern of new and regular customers, statistically there is no difference in spending nature in between of them. Hence for the business, high revenue generation is through frequency based not spending based. Hence the business should shift its focus from increase the basket size to first to second visit conversion program.